



Power evolution prediction of bidirectional Raman amplified WDM system based on PINN

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Abstract: We propose using physical-informed neural network (PINN) for power evolution prediction in bidirectional Raman amplified WDM systems with Rayleigh backscattering (RBS). Unlike models based on data-driven machine learning, PINN can be effectively trained without preparing a large amount of data in advance and can learn the potential rules of power evolution. Compared to previous applications of PINN in power prediction, our model considers bidirectional Raman pumping and RBS, which is more practical. We experimentally demonstrate power evolution prediction of 200 km bidirectional Raman amplified wavelength-division multiplexed (WDM) system with 47 channels and 8 pumps using PINN. The maximum prediction error of PINN compared to experimental results is only 0.38 dB, demonstrating great potential for application in power evolution prediction. The power evolution predicted by PINN shows good agreement with the results simulated by traditional numerical method, but its efficiency is more suitable for establishing models and calculating noise, providing convenience for subsequent power configuration optimization.

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1. Introduction

With the rapid development of the Internet and mobile apps, communication technology is continuously updating and the capacity demand for fiber-optic communication system is constantly increasing [1–3]. Adopting wavelength-division multiplexed (WDM) system with a wider transmission band to load more channels instead of laying new fiber optic cables is an economical solution [4]. Raman amplification is a key technology for increasing the system capacity and transmission distance, as it has a wider and adjustable gain spectrum compared to the erbium-doped fiber amplifiers (EDFAs) [5]. The spectral flexibility of Raman amplification allows the gain spectrum to be shaped by combining different pump wavelengths and regulating pump power [6]. Using this broad-band pumping approach, amplifiers with gain bandwidths greater than 100 nm have been demonstrated [7].

However, the effect of stimulated Raman scattering (SRS) does not only occur between the pump and signal. Energy will transfer from the high-frequency component to the low-frequency component, which leads to strong Raman interactions between pumps and between signals. This energy transfer not only changes the power and signal-to-noise ratio of different channels, but also impacts the distortion arising from Kerr nonlinearity, as it changes the power evolution of each channel. When designing such broad-band Raman amplifiers, the inter-channel stimulated Raman scattering (ISRS) becomes significant and it must be considered [8–10].

Optimization algorithms such as evolutionary algorithms [11–14], annealing algorithms [15] and particle swarm optimization [16] have been proposed to get a flat gain in Raman amplification system, which are usually based on accurate forward models that can provide a mapping relationship between power configuration and Raman gain. In order to accurately evaluate the gain spectrum, a high-fidelity power evolution model of pumps and signals is in need. The Raman power coupling equation in the form of partial differential equations (PDEs) is usually used to characterize the power evolution of WDM pumps and signals, which is usually difficult to be solved analytically [17]. Numerical methods such as the Runge-Kutta method are often used to solve the Raman equations. In general, it is complex and time consuming as it requires the solution of a two-point boundary condition problem as long as the backward Raman pumps are involved. The power of the backward Raman pumps at the transmitting end ($z = 0$) is unknown and needs to be iteratively calculated using shooting methods, etc., and in some cases the solution may not converge. If bidirectional Raman pumping scheme is used, the Rayleigh backscattering (RBS) of the forward pumps will become a part of the backward pumps, further amplifying the signal, and vice versa [18,19]. This further complicates the problem. Another solution is to use machine learning (ML) [20]. Most ML techniques build an approximate model in a data-driven manner. The advantage is that it can solve the equations in a shorter time in the absence of prior knowledge. Black-box neural network with fiber agnostic training can provide either forward modelling [21] or reverse system design [22]. However, these data-driven methods rely on the quality and quantity of the data, which makes the preliminary preparation more complicated. Besides, it is prone to underfitting or overfitting, and often lack generalization, which is not expected [23–25]. Other solution like differentiable PDE-based systems [26,27] adopt certain approximations in the calculation process, and can only obtain the output power at the receiver rather than the power evolution along the fiber link, which is unfavorable for evaluating impacts of nonlinearity.

Recently, physics-informed neural network (PINN) that combines the advantages of ML with physics has been arousing widespread interest in solving different kinds of PDEs [28–31]. Compared to traditional numerical methods, PINN is efficient and has no special requirements about how to set the initial values. On the other hand, compared to data-driven ML techniques, PINN follows physical laws and satisfies various boundary conditions or other constraint conditions, providing an interpretable and generalized model. In addition, PINN does not require too much data because it is an unsupervised approach. It can provide accurate and reasonable predictions even for extrapolatory scenarios that do not exist in the training data. Moreover, PINN provides power evolution along the entire fiber link, which can help calculate the impact of nonlinearity.

In this paper, we propose using PINN for power evolution prediction in bidirectional Raman amplified WDM systems with RBS, and prove the accuracy of PINN through experiments and numerical simulations. This paper will be divided into four main parts. A brief background of power evolution of Raman amplified WDM system and comparison of numerical method, data-driven ML method and PINN are introduced in the introduction. Then we introduce the mathematical expression of SRS in Raman amplified WDM systems and the structures and configurations of PINN in detail in the second part. In the third part, we give our experimental setup and compare the results of experiments, numerical simulation and PINN. Finally, conclusions and future research work will be illustrated in the last part.

2. Fundamentals of Raman amplifier and PINN

2.1. General Raman equations

Ignoring the influence of amplified spontaneous emission (ASE) noise, the power evolution of WDM pumps and signals follows the propagation equations. The power $P(z, \nu)$ of the channel

with frequency ν at distance z follows the equation [32,33]:

$$\frac{dP^\pm(z, \nu)}{dz} = \mp \alpha(\nu) P^\pm(z, \nu) \pm \kappa(\nu) P^\mp(z, \nu) \pm P^\pm(z, \nu) \sum_{\varsigma} C_R(\varsigma, \nu) [P^\pm(z, \varsigma) + P^\mp(z, \varsigma)] \quad (1)$$

where, ν and ς are the optical frequencies, the + and - symbols denote the propagation directions, α is the fiber loss, κ is the Rayleigh backscattering coefficient, $C_R(\varsigma, \nu)$ is the Raman coupling coefficient between frequency ς and ν components:

$$C_R(\varsigma, \nu) = \begin{cases} \frac{g_R(\varsigma, \nu)}{K_{eff} A_{eff}} & \varsigma > \nu \\ 0 & \varsigma = \nu \\ -\frac{\nu}{\varsigma} \frac{g_R(\nu, \varsigma)}{K_{eff} A_{eff}} & \varsigma < \nu \end{cases} \quad (2)$$

where $g_R(\varsigma, \nu)$ represents the Raman gain coefficient between frequency components ν and ς , A_{eff} is the effective area of the optical fiber, K_{eff} is the polarization factor between pump and Stokes signals, $K_{eff} = 2$ when the polarization state of pump light and signal light is random.

Define matrix $\mathbf{G}$ to characterize the Raman coupling between various frequency components in the entire WDM system:

$$\mathbf{G} = \begin{bmatrix} 0 & C_R(\nu_2, \nu_1) & \cdots & C_R(\nu_N, \nu_1) \\ C_R(\nu_1, \nu_2) & 0 & \cdots & C_R(\nu_N, \nu_2) \\ \vdots & \vdots & \ddots & \vdots \\ C_R(\nu_1, \nu_N) & C_R(\nu_2, \nu_N) & \cdots & 0 \end{bmatrix} \quad (3)$$

where N is the total number of channels including all pumps and signals.

As long as the power evolution of the signals and pumps is calculated, the corresponding amplified spontaneous emission (ASE) noise power $P_{ASE}(z, \nu)$ can be obtained using the following equation:

$$\frac{dP_{ASE}^\pm(z, \nu)}{dz} = \left[-\alpha(\nu) + \sum_{\nu < \varsigma} C_R(\varsigma, \nu) [P^+(z, \varsigma) + P^-(z, \varsigma)] \right] P_{ASE}^\pm(z, \nu) + \kappa(\nu) P_{ASE}^\mp(z, \nu) + 2h\nu \Delta\nu \sum_{\nu < \varsigma} C_R(\varsigma, \nu) [P^+(z, \varsigma) + P^-(z, \varsigma)] \left[1 + \frac{1}{e^{h(\varsigma - \nu)/kT} - 1} \right] \quad (4)$$

where h is the Planck's constant, k is the Boltzmann constant and T is the Kelvin temperature. $\Delta\nu$ is the noise bandwidth.

2.2. Principle of PINN

The structure of PINN based Raman amplified WDM system is depicted in Fig. 1. It can be divided into two parts: the main NN and the loss calculation. The model based on PINN views fiber transmission as a PDE solving problem, so input power of pumps and signals should be involved as a constraint in the loss calculation from the beginning. The NN first constructs a mapping relationship between inputs and outputs, and then the mean square error (MSE) term is calculated based on the power sequence output by NN to form the loss function. Next, the parameters in the NN is optimized through backpropagation to minimize the loss function. Finally, the NN is completely trained, and can provide accurate mapping relationships between inputs and outputs.

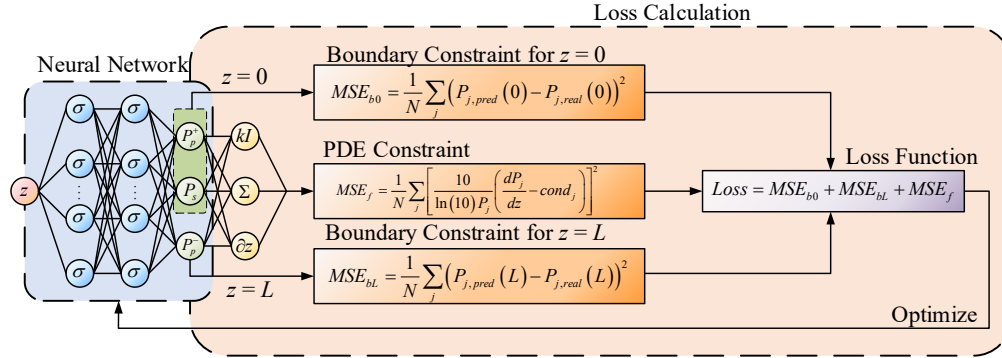


Fig. 1. Frame of the PINN. σ denote the parameters of NNs to be optimized. The actual output of NN is a power array $\mathbf{P}(z) = \begin{bmatrix} P_p^+(z, \nu_1) & \dots & P_p^+(z, \nu_N) & P_p^-(z, \nu_1) & \dots & P_p^-(z, \nu_N) & P_s(z, \nu_1) & \dots & P_s(z, \nu_N) \end{bmatrix}^T$

A multi-layer fully-connected neural network is used in this work, which include 1 input, 4 hidden layers with 38 neurons in each layer, and 55 outputs (corresponding to 47 channels and 8 pumps). The activation function is tanh. The input is the vector of spatial distance z which takes values from 0 to L (L is the fiber length), and the output is a 2D matrix of the power $\mathbf{P}(z) = \begin{bmatrix} P(z, \nu_1) & P(z, \nu_2) & \dots & P(z, \nu_N) \end{bmatrix}^T$ at corresponding position and frequency. To avoid the impact of magnitude differences between pump and signal power, we represent the output power $\mathbf{P}(z)$ in dBm units. This operation is necessary since the NN can effectively map the actual inputs to the range where the model can be trained at the best efficiency.

The loss function consists of two parts: boundary condition and PDE constraints. The boundary condition is the input power of forward pumps as well as signals at $z = 0$, and the input power of backward pumps at $z = L$. MSEs are used to penalize the results that do not satisfy the constraints:

$$MSE_{b0} = \frac{1}{N} \sum_j (P_{j_dB,pred}(0) - P_{j_dB,real}(0))^2 \quad (5)$$

$$MSE_{bL} = \frac{1}{N} \sum_j (P_{j_dB,pred}(L) - P_{j_dB,real}(L))^2 \quad (6)$$

where subscripts $b0$ and bL refer to boundary conditions at $z = 0$ and $z = L$, respectively, and subscripts $pred$ and $real$ correspond to the PINN predicted values and the actual input of the system. j refer to the j -th channel, and dB represents the value in dB as the unit.

The PDE residual of Eq. (1) is embedded to constrain the outputs of NNs in an unsupervised way. The PDE function $cond$ is a matrix with dimensions the same as the number of transmission channels. In order to match the magnitude of the PDE loss function with the boundary conditions mentioned before, we calculate the PDE loss in logarithmic form (in dB). This operation is beneficial for keeping the derivative of the function within a small range. If linear unit is used, the derivative is prone to fluctuations on the order of magnitude. The PDE loss is then expressed as:

$$MSE_f = \frac{1}{N} \sum_j \left[\frac{10}{\ln(10)P_{j_lin}} \left(\frac{dP_{j_lin}}{dz} - cond_j \right) \right]^2 \quad (7)$$

$$cond = [-\alpha + \mathbf{G} \times \mathbf{P}_{lin}(z)]\mathbf{P}_{lin}(z) + \kappa \mathbf{P}_{lin}^-(z) \quad (8)$$

where $\alpha = [\alpha_1 \ \alpha_2 \ \dots \ \alpha_N]^T$ is the array of attenuation coefficients of all channels (for backward pumps with opposite propagation directions, a negative sign needs to be added), $\kappa = [\kappa_1 \ \kappa_2 \ \dots \ \kappa_N]^T$ is the array of the Rayleigh backscattering coefficients of all channels, similarly. The subscript j represents the j -th item in the array. It is worth noting that the unit of P_{lin} in Eq. (7) and Eq. (8) is linear (mW) so we marked *lin* in the subscript.

The loss function can be expressed as

$$Loss = MSE_{b0} + MSE_{bL} + MSE_f \quad (9)$$

In the training of PINN, the widely-utilized Adams algorithm based on gradient descend is adopted. 2,000 coordinates randomly sampled from 0 to L are used as inputs in each iteration. The differential terms in the PDE loss function can be calculated efficiently using the automatic differentiation built in deep learning libraries. To improve convergence speed, during the initial iteration stage, a fixed coordinate grid can be used as input, and replacing differentiation with small-step differences as Eq. (10) illustrates can further improve computational efficiency.

$$\frac{dP(z)}{dz} = \frac{P(z + \Delta z) - P(z)}{\Delta z} \quad (10)$$

where Δz is the microelements of distance and takes a value of 0.01 km in this work. By this way we can save the process of solving derivative expressions and avoid overly complex “expression explosions”, while the result has a considerable accuracy.

3. Experimental transmission and result verification

The experimental setup is depicted in Fig. 2. A total of 47 channels from 196 THz to 191.4 THz are loaded, of which 4 channels are real optical signals sent from commercial 400 G optical modules, and the other 43 waves are channelised ASE signals generated by optical amplifier. The total input signal power into the forward RFA is 17 dBm. The input spectrum is measured at point A and marked in Fig. 2(b). The transmission link is composed of 200 km G.652 fiber with a total attenuation of 38.1 dB. We collect spectrum at point B as output. The gain of the pre-amplifier (PA) is about 15 dB and the noise figure is about 5 dB. We also measure the optical signal-noise-ratio (OSNR) of the four real channels at the receiver end.

The configuration of the Raman pumps is shown in Table 1. Long-wavelength pumps are with lower power to reduce ISRS, while the power of forward pumps is not fully loaded in order to suppress the influence of nonlinearity. The dispersion coefficient β_2 is $-21.7 \text{ ps}^2/\text{km}$, the nonlinear coefficient γ is $1.2 \text{ (W}\cdot\text{km)}^{-1}$, and the RBS coefficient κ is $5 \times 10^{-5} \text{ km}^{-1}$ at 1550 nm.

Table 1. Configuration of the Raman Pumps

Wavelength (nm)	Forward Pump Power (mW)	Backward Pump Power (mW)	Attenuation (dB/km)
1425nm	200	250	0.2710
1435nm	150	250	0.2580
1455nm	100	250	0.2360
1465nm	100	250	0.2300

Restricted by experimental conditions, the power evolution of the pumps and signals along the fiber link is not measured by OTDR experimentally. To verify the accuracy of PINN prediction, we simulate the experimental system using numerical methods and PINN, and compare the power evolution of the pumps and signals calculated by the two methods. The comparison of evolution of Raman pumps is shown in Fig. 3. Short-wavelength pumps experience greater attenuation, partly due to their larger attenuation coefficients, and partly due to the SRS consumption by

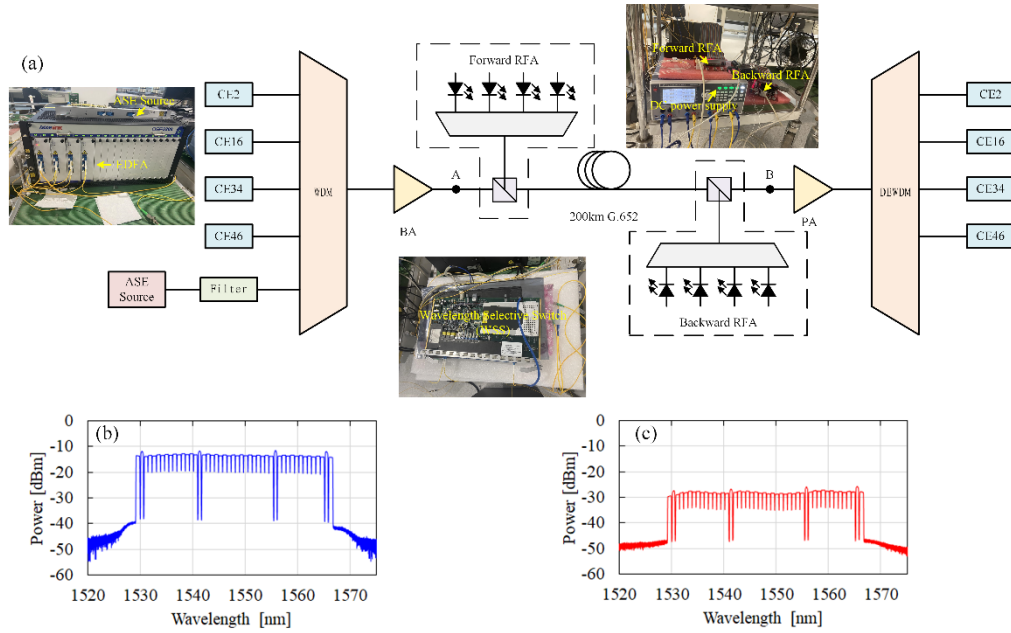


Fig. 2. Transmission experiment setup and measured spectrum. (a) Experiment setup; (b) The input spectrum measured at point A (the output side of BA); (c) The output spectrum measured at point B (the output side of the backward RFA).

other long-wavelength pumps. The forward pump power has increased near the end of the fiber, mainly due to the RBS of the backward pumps. The backward pumps also exhibit a similar power increase near the front side of the fiber due to the RBS of the forward pumps. Overall, the evolution of pump power predicted by PINN is not significantly different from that calculated by numerical methods, with only a certain difference in the 1425 nm backward pump, which is acceptable.

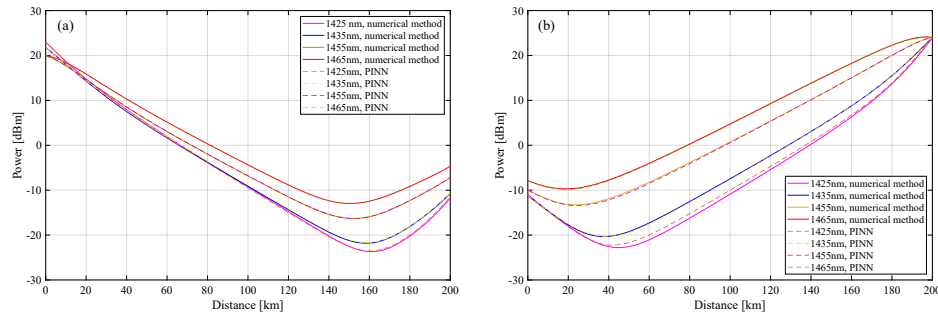


Fig. 3. Evolution of pumps calculated by numerical methods and PINN. (a) Forward pumps; (b) Backward pumps

We also compared the output signal spectrum power in Fig. 4. Short-wavelength channels are with significantly lower power while long-wavelength channels are with significantly higher power, due to the wavelength dependence of the attenuation curve and the power transfer caused by ISRS. Compared with the experimental results, the maximum prediction error of the numerical method is 0.42 dB, and the root mean square error (RMSE) is 0.22 dB, while the maximum

prediction error of PINN is 0.38 dB, and the RMSE is 0.35 dB. The channel with the largest error predicted by PINN is the one with the longest wavelength, so we present the power evolution of this channel in Fig. 5. It can be seen that the error is mainly reflected in the latter half of the optical fiber, which is mainly due to the relatively inaccurate prediction of the backward pump power at the end of the optical fiber. Even so, the maximum error between PINN and the numerical method in predicting signal power is only 0.28 dB, which is completely within an acceptable range. The PINN structure consists of 366,685 multipliers in total, while the numerical method requires 21,221 calls to the PDE equation during the iteration process for a fiber length with 200 km, and each call requires at least 6,105 multiplication operations. The number of operations for numerical methods will further increase with increasing distance and mesh refinement. In summary, PINN exhibits low complexity in computational efficiency once the model is trained.

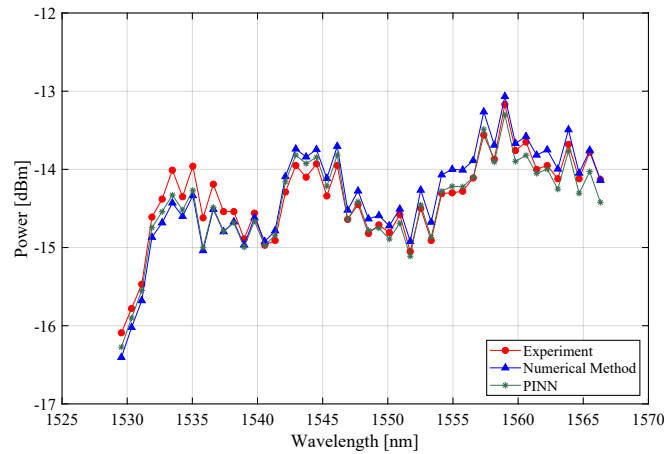


Fig. 4. Comparison of output spectrum of experiments, numerical methods and PINN.

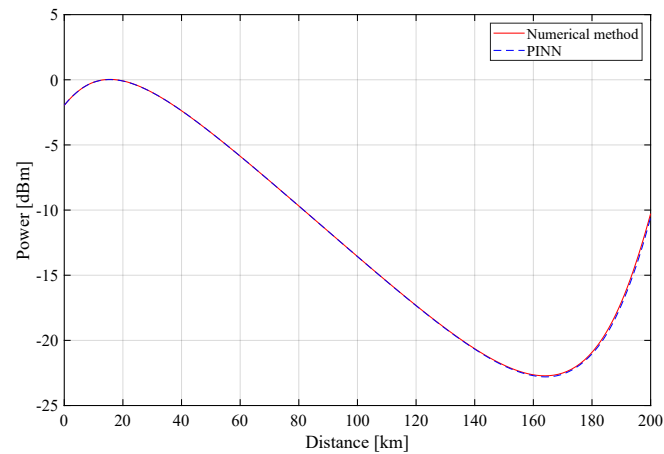


Fig. 5. Evolution of signal calculated by numerical methods and PINN.

Although the proposed PINN model does not provide noise power directly, we can substitute the predicted power evolution of PINN into Eq. (4) to calculate the ASE power and OSNR. The noise power considered in calculating OSNR includes the ASE accumulated along the link as

shown in Eq. (4), the nonlinear interference (NLI) illustrated by [34, Eq. (2)], and the ASE generated by PA, where the first two are indirectly derived from power evolution. We compare the OSNR of the 4 real channels between experiment and PINN, and the results are listed in Table 2. The OSNR of the 4 real channels is measured by calculating the ratio of the integral power within the corresponding frequency interval displayed on the optical spectrum analyzer with the channel on and off. Just like the output power, short-wavelength channels are with significantly lower OSNR while long-wavelength channels are with significantly higher OSNR, due to the power transfer caused by ISRS. The prediction results of PINN shows good agreement with the experiment, and the maximum absolute error is only 0.42 dB. The prediction results of channels 34 and 46 are more accurate, with an error less than 0.1 dB, indicating that the power evolution predicted using PINN has high accuracy.

Table 2. Comparison of OSNR

Channel Number	OSNR (dB)		
	Experiment	PINN	Error
2	26.45	26.84	0.39
16	27.26	26.84	-0.42
34	27.49	27.55	0.06
46	28.04	28.06	0.02

To verify the generalization of the model, we simulated scenarios with the same power configuration and different fiber lengths from 160 km to 240 km. Due to limitations in experimental conditions, we did not conduct further experimental tests, and these results were all derived from simulations. The comparison of output spectrum calculated by numerical method and PINN is shown in Fig. 6. It should be noted that changing the system (including power configuration, fiber length, etc.) will make the original PINN model no longer applicable and the model must be retrained with new parameters. The results show that under different fiber lengths, the max absolute error between the output spectrum predicted by PINN and the numerical method is less than 0.5 dB, indicating that PINN can provide accurate power prediction for different scenarios.

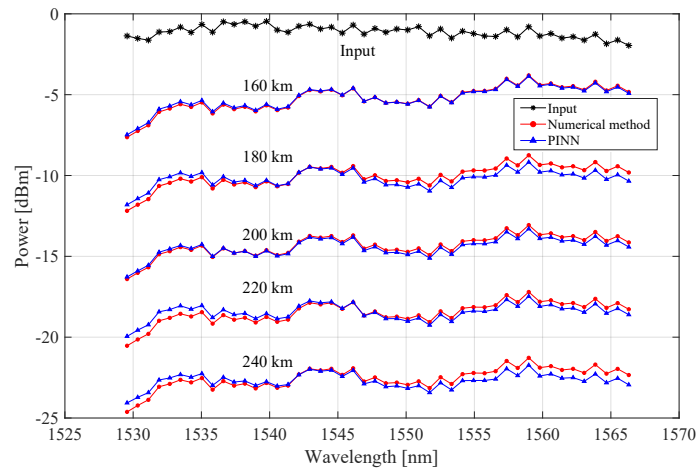


Fig. 6. The output power spectrum with different distances calculated by numerical methods and PINN.

4. Conclusion

In this paper, a PINN-based power evolution model for bidirectional Raman amplified WDM system with RBS is proposed. The proposed method is more efficient and easier to converge compared to numerical simulation, and does not require a large amount of training data compared to data-driven ML. The proposed model considers bidirectional Raman pumping and RBS, which is closer to practical scenarios compared to previous PINN applications. And we eliminate the numerical magnitude difference between pump and signal power by using a logarithmic loss function. We analyzed the principle of PINN and compared its accuracy with experiments and numerical simulations. In a 200 km bidirectional Raman amplified WDM system with 47 channels and 8 pumps using PINN, the maximum power prediction error of PINN compared to experimental results is only 0.38 dB, and the maximum OSNR prediction error is 0.42 dB. PINN combines the advantage of numerical methods and ML, demonstrating great potential in power prediction. Further applications in fiber optics like parameters identification of fiber characteristics and system performance optimization may be achieved by PINN, as it is a prospective computing tool.

Disclosures. The authors declare that there are no conflicts of interest related to this article.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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